

IoT protocols and network standards

There are many IoT protocols available for different applications and requirements. However, each has its own advantages and disadvantages for different IoT scenarios. Some of the most widely used IoT protocols are discussed here.

Message queue telemetry transport (MQTT) protocol

This open source publish/subscribe messaging transport protocol is very lightweight and ideal for connecting small devices to constrained networks. It was designed to work in low-bandwidth conditions, such as sensors and mobile devices, on unreliable networks. This capability makes it the most preferred protocol for connecting devices with small code footprint, as well as for wireless networks with varying levels of latency due to bandwidth constraints or unreliable connections. It works on top of transmission control protocol/Internet protocol (TCP/IP) to provide reliable delivery of data. MQTT has three main components:

- Subscriber
- Publisher
- Broker

The basic workflow of this protocol involves the publisher being responsible for generating and transmitting information to subscribers through a broker. The main function of the broker is to ensure security by checking the authorisation of subscribers and publishers. This protocol is preferred for IoT devices because it provides well-organised information routing functions to small, cheap, low memory, power devices and low bandwidth networks. To ensure message reliability, MQTT uses three levels of quality of service (QoS).

- *QoS0 (At most once)*: This is the least reliable but the quickest mode. The message is sent without any confirmation received.
- *QoS1 (At least once)*: Ensures that the message is delivered at least

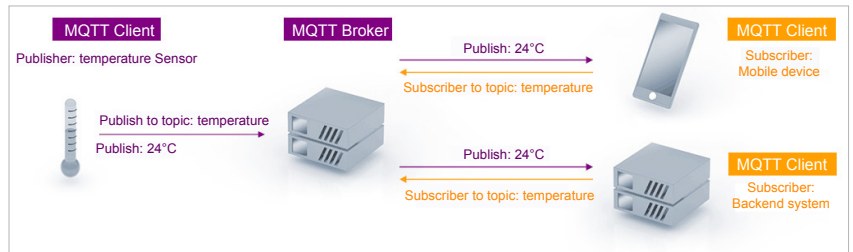


Figure 3: MQTT publish/subscribe architecture (Reference: <https://mqtt.org/>)

once, even if duplicate messages may be received.

- *QoS2 (Exactly once)*: This is the most reliable but also the most bandwidth-consuming mode. Duplicate messages are controlled to ensure that it gets delivered only once.

MQTT is a bi-directional communication protocol, where the clients can produce and consume data by publishing messages and subscribing to topics. Two-way communication enables IoT devices to send sensor data and simultaneously receive

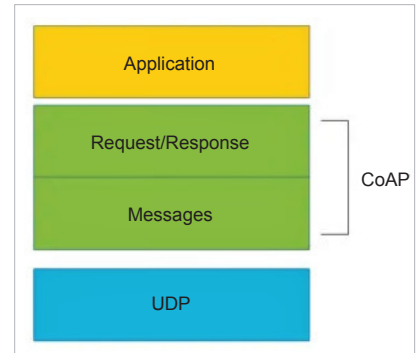


Figure 4: CoAP messaging model (Reference: <https://dzone.com/articles/coap-protocol-step-by-step-guide/>)

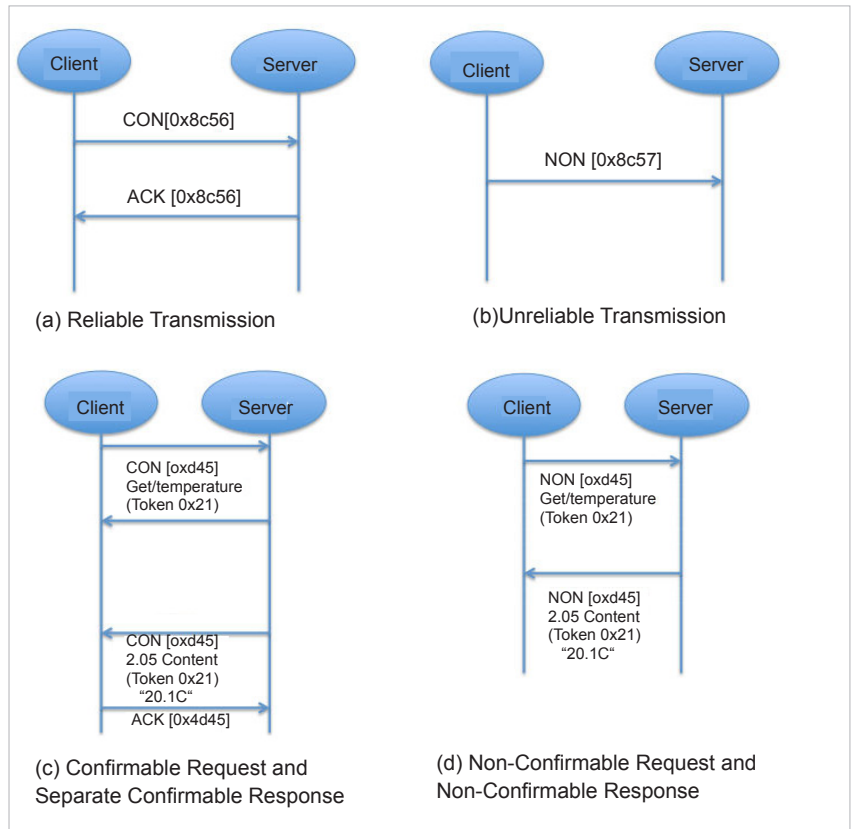


Figure 5: CoAP message flows (<https://devopedia.org/constrained-application-protocol/>)